

Lecture 37: SMRs on our Horizon: BWRX, Sodium, Xe-100, and Kairos

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: Scaling and Strategy

Over the course of the semester we have analyzed the fundamental physics of PWR, BWR, CANDU, HTGR, and LMFR systems. Today, we synthesize these into the four specific Small Modular Reactor (SMR) designs that may well be employed in our region in the near future. We move from "How they work" to "Why they are constrained." None of these reactors are currently operational anywhere in the world, but that is shortly going to change.

1 The BWR: GE-Hitachi BWRX-300

The Proponent: Proposed for the Rockport Generating Station (Spencer County, IN) by Indiana Michigan Power (I&M/AEP).

The Pedigree: The 10th generation of GE Boiling Water Reactor technology. It is a "right-sized" version of the 1,520 MWe ESBWR (Economic Simplified BWR).

Current Status: Lead unit under construction at Darlington (Ontario) with a 2028-2029 target. I&M has utilized a \$50M DOE grant to begin the Early Site Permit (ESP) process for the Rockport, Indiana site on the Ohio River as of early 2026.

1.1 Power, Scale, and the Cook Benchmark

To understand the "Small" in SMR, we compare it to local Generation II assets:

- **Rating:** The BWRX-300 is rated at **300 MWe** (electrical) and **870 MWt** (thermal).
- **The Benchmark:** The nearby **Donald C. Cook** plant (Bridgman, MI) operates two Westinghouse PWRs at $\approx 1,030$ MWe and $\approx 1,077$ MWe respectively.
- **The Displacement:** A single BWRX-300 provides $\approx 30\%$ of the power of one Cook unit. I&M's strategy at Rockport involves a "twin-pack" (600 MWe) to replace retiring coal capacity while utilizing existing 765 kV transmission infrastructure.

1.2 Design: Natural Circulation and the "Direct Isolation" Strategy

The BWRX-300 achieves 100% power via **Natural Circulation**, but its most critical safety innovation is the management of the Loss of Coolant Accident (LOCA) pressure spike.

- **The Driving Head:** Utilizing an internal "**Chimney**" ($\approx 10\text{m}$ tall) above the core, the density gradient between the boiling two-phase mixture in the chimney and the subcooled liquid in the downcomer drives the flow without mechanical pumps.
- **The Direct Isolation Valve (DIV):** In traditional BWRs, isolation valves are located downstream in the piping. A "Large Break" between the vessel and the valve allows the entire reactor inventory to depressurize into containment.
- **Nozzle-Mounted Isolation:** The BWRX-300 mounts its High-Pressure Isolation Valves **directly to the RPV nozzles**. If a steam pipe breaks, the DIV closes immediately at the vessel boundary.
- **Energy Limitation:** By "locking" the high-energy steam inside the vessel, the mass and energy released into the silo during a break is reduced by orders of magnitude compared to a PWR or ESBWR. This allows the containment volume to be reduced to a compact, **Underground Silo** without risking over-pressurization of the structure.

1.3 Containment: The Underground Silo and Pressure Suppression

Because the DIVs limit the total steam release, the "Containment" can shift from a large atmospheric dome to a high-pressure **Pressure Suppression Pool** strategy.

- **Quenching the Pulse:** Any steam that does escape before DIV closure is immediately routed through "Downcomers" into a circular pool of water surrounding the base of the reactor. The steam rapidly condenses, "quenching" the pressure spike before it can stress the silo walls.
- **Soil-Structure Interaction:** The **Underground Silo** ($\approx 35\text{m}$ deep) uses the surrounding earth as a secondary structural member. Unlike an above-ground dome which must resist internal pressure through the tensile strength of reinforced concrete alone, the silo walls transfer the load directly into the surrounding geological formation.

1.4 Safety Margins: What if the DIV Fails?

The "Defense-in-Depth" strategy ensures core cooling even in the event of a Direct Isolation Valve failure:

- **Redundancy:** Each RPV nozzle is equipped with **two independent valves in series**. A mechanical failure in one does not prevent isolation by the second.
- **The Isolation Condenser (ICS):** A 100% passive heat exchanger submerged in an elevated water pool. Steam is condensed and gravity-fed back into the core, providing **7 days of cooling** without electricity or pumps.
- **Pressure Suppression:** If isolation fails, the **Suppression Pool** at the base of the silo acts as a heat sink, condensing steam via downcomers to prevent over-pressurization of the underground structure.
- **Gravity-Driven Cooling (GDCS):** In a depressurization event, elevated emergency pools provide a passive "flooding" of the core via gravity, ensuring the fuel remains submerged regardless of valve status.

1.5 The Constraint: The Water-Energy Nexus

As a Light Water Reactor (LWR), the BWRX-300 is tethered to a massive heat sink for its Rankine cycle.

- **Location Sensitivity:** This design is ideal for the **Ohio River** (Rockport), which provides the consistent flow-rate required for 570 MW of rejected thermal energy.
- **Regional Limits:** Inland "Dry" sites are likely unsuitable for this technology. The St. Joseph River flow-rate, for example, is insufficient for a plant of this scale without thermal impacts or evaporative losses that would threaten the local water table.

2 The LMFR: TerraPower Natrium

The Proponent: TerraPower (Founded by Bill Gates).

The Physics: A Liquid Metal Fast Reactor (LMFR)—specifically a Sodium-Cooled Fast Reactor (SFR)—using metallic HALEU fuel, coupled with a molten salt energy storage system.

Current Status: Lead project in Kemmerer, WY began non-nuclear construction in 2024. In March 2026, the NRC officially issued its construction permit—the first commercial non-light-water reactor construction permit approved in the U.S. in over 40 years.

2.1 Neutronics and Stability

- **The Fast Spectrum Edge:** Fast reactors can exhibit a **positive sodium void coefficient** in portions of the core, meaning that local coolant voiding can increase reactivity if not countered by other feedback mechanisms.
- **Passive Feedback:** To be "walk-away safe," the Natrium core is designed with high **Thermal Expansion Feedback**. As the core gets hot, the metallic fuel slugs and grid structures physically expand. This lowers the fuel density and increases neutron leakage, shutting down the chain reaction faster than the positive voiding effect can ramp it up. The overall temperature coefficient is negative, as required by NRC regulations.

2.2 The Guard Vessel Strategy

Natrium operates at near-atmospheric pressure. There is no heavy "Pressure Vessel" to explode or blow down. Instead, it relies on a **Guard Vessel** (a tank-in-a-tank design). If the primary reactor vessel springs a leak, the guard vessel catches the sodium, ensuring the core remains safely submerged and capable of passive decay heat removal.

2.3 The Primary Vulnerabilities: Chemistry and Opacity

As a coolant, liquid sodium brings unique engineering challenges:

- **The Chemical Threat:** Sodium burns when exposed to oxygen (air) and reacts violently, producing explosive hydrogen gas, when exposed to water.
- **The Defense (The Three-Loop Isolation):** To keep high-pressure steam as far from the radioactive core as possible, Natrium utilizes a cascaded heat-transfer system:

1. *Primary Sodium Pool:* Radioactive sodium directly cools the core inside the reactor vessel.
2. *Secondary Sodium Loop:* A non-radioactive intermediate sodium loop extracts heat from the primary pool and carries it out of the nuclear island.
3. *Molten Salt Loop (Solar Salt):* The secondary sodium transfers its heat to a molten nitrate salt loop via a heat exchanger. While sodium and nitrate salt are chemically reactive, this interface operates at near-atmospheric pressure, preventing the catastrophic, explosive blowdown that a high-pressure steam rupture would cause.
4. *Steam Cycle:* The hot solar salt travels to the thermal reservoirs in the Energy Island, where it is finally used to boil water into high-pressure steam.

By the time the thermal energy reaches water, it is physically and chemically isolated from the nuclear core.

- **Opacity:** Because sodium is a visually opaque liquid metal, operators cannot "see" the fuel or internal mechanisms. Maintenance and refueling require specialized ultrasonic sensors and blind robotics.

2.4 The "Energy Island": Nuclear as a Peaker Plant

Natrium's most unique feature is how it completely decouples the nuclear reactor from the steam turbine to solve the problem of intermittent renewable grids.

- The reactor constantly runs at 100% capacity, generating a steady baseline of 840 MWt (producing a nominal 345 MWe).
- Instead of boiling water immediately, the reactor dumps this heat into massive vats of molten nitrate salt (similar to concentrated solar power plants).
- The steam turbine draws its heat from these salt tanks, not the reactor. This allows the plant to throttle its electrical output between **100 MWe** (when wind/solar are abundant) and a boosted **500 MWe** (acting as a "peaker" plant when grid demand spikes), all without ever moving the reactor's control rods.

3 HTGR: X-Energy Xe-100

The Proponent: Amazon.

The Physics: A High-Temperature Gas-Cooled Reactor (HTGR) using TRISO fuel pebbles and Helium coolant.

Current Status: The first four-unit Xe-100 plant (80 MW each) is planned for Dow Chemical's Seadrift, TX site, with early site preparation and licensing activities underway. Amazon has directly invested in X-energy to fund follow-on deployments, specifically targeting the **Cascade Advanced Energy Facility** in Washington State (in partnership with Energy Northwest), with initial plans for four reactors (320 MW) scaling up to twelve (960 MW) to power AWS data centers.

3.1 Core Geometry & The "Hot Spot" Ghost

- **Cylindrical Geometry ("Pencil" Core):** To minimize heat transfer issues, the Xe-100 uses a tall, narrow, cylinder of pebbles (heavily influenced by the German HTR-Modul and

Chinese HTR-PM). This "pencil" shape ensures that even a pebble stuck dead-center is physically close enough to the outer wall to passively dump decay heat via conduction.

- **The Hot Spot Issue (AVR Experience):** Data from the original German AVR reactor (the "Melt Wire" data) proved that a pebble bed is not uniform to fluid flow. As in the flow through any packed bed, helium takes the path of least resistance, potentially bypassing any tightly packed region of the bed. In the AVR, this created hidden, low-flow "hot spots" which led to unexpectedly high pebble temperatures.
- **The Failure Point:** If localized hot spots combine with a reactivity spike or loss of flow, pebbles could theoretically approach 1600°C to 1650°C. This is the critical limit where the Silicon Carbide (SiC) pressure layer in the TRISO fuel begins to thermally degrade, leaking fission products into the primary helium loop. The narrow core geometry is X-energy's primary physical defense against reaching this temperature.

3.2 The Primary Vulnerability: Water Ingress

- **The Pressure Differential:** In the Xe-100, the primary Helium coolant operates at roughly **7 MPa** (approx. 1,000 psi). However, the secondary loop (the Steam Generator) produces superheated steam at **16.5 MPa** (approx. 2,400 psi). Because the steam pressure is more than double the helium pressure, a ruptured tube in the heat exchanger guarantees that high-pressure water and steam will be driven *into* the reactor core, rather than helium safely leaking out.
- **Heat Exchanger Design:** To reduce size the heat exchangers consist of a shell (helium) and coiled steam generating tube design. Under some conditions the tube coils act as a large spring with the helium flowing at high velocities perpendicular to the coils. This introduces the possibility of flow induced resonance and vibration which can lead to progressive weakening and rupture of the tubes. Simulations indicate that structural strengthening can reduce the magnitude of this effect, but fabrication defects (e.g., the cause of the Pickering accident) remain a concern.
- **Regulatory Focus on Flow-Induced Vibration:** Publicly available NRC audit materials and requests for additional information (RAIs) indicate that the behavior of the helical-coil steam generator under cross-flow conditions remains an area of active regulatory scrutiny. In particular, the NRC has requested quantitative modeling of flow-induced vibration and structural response. This request reflects the need to evaluate coupled fluid-structure interaction effects, including the potential for resonance, fatigue, and long-term degradation. Such analyses must account not only for nominal design conditions but also for plausible variations in fabrication tolerances, which can influence stiffness, boundary conditions, and resonant frequencies.
- **The Chemical Threat (Steam-Graphite Reaction):** When steam hits the heavily heated graphite pebbles and reflector blocks, it triggers a rapid chemical oxidation. The water reacts with the carbon to produce **Hydrogen** and **Carbon Monoxide** gases. This not only physically degrades the structural graphite but also fills the primary loop with a highly pressurized, flammable, and toxic mixture.
- **The Defenses:**

- *Detection and Isolation:* The primary loop is equipped with moisture monitors. If humidity spikes, the system automatically scrams the reactor, isolates the Steam Generator, and dumps the secondary water inventory to physically sever the source of the ingress.
- *Pressure Relief:* If the reaction produces gas faster than the isolation valves can respond, the reactor pressure vessel relies on mechanical relief valves. These valves vent the volatile mixture of helium, steam, hydrogen, and carbon monoxide directly into the reactor building to prevent the primary steel vessel from catastrophically bursting. If the pressure in the building becomes too high, it is designed to then vent this plume through filters into the atmosphere.

3.3 Functional Containment and TRISO Performance

Unlike traditional light-water reactors, the Xe-100 has no high-pressure reinforced concrete dome. It relies entirely on **Functional Containment**, an approach where the TRISO fuel particle itself acts as the primary barrier against radiological release. Instead of a macroscopic physical barrier catching a release after it happens, functional containment depends on the microscopic performance of the Silicon Carbide (SiC) layer under both normal and accident conditions:

- **Under 1200°C (Normal Operation):** Fission products are almost entirely retained. Only highly mobile trace isotopes can slowly diffuse through the completely intact SiC pressure-bearing layer.
- **Up to 1600°C (Design Basis Accident Limit):** The TRISO layers remain structurally sound. The SiC layer continues to act as a highly effective barrier, securely retaining the most dangerous radiological hazards like Cesium (Cs-137) and Strontium (Sr-90), as well as noble gases like Krypton and Xenon, even after hundreds of hours.
- **Above 1600°C to 1800°C (Beyond-Design Limit):** At these extreme temperatures, the SiC layer experiences progressive thermal degradation. Cesium and Strontium begin escaping, and if the particle physically ruptures, highly volatile noble gases will vent into the surrounding graphite.

Historically, in the 1980s German gas-cooled reactors, TRISO manufacturing defects and irradiation-induced failures hovered around 1×10^{-4} to 5×10^{-4} . However, modern advanced coating techniques (like those in the US DOE AGR Program) have dropped the failure rate to $< 2 \times 10^{-5}$, with recent landmark tests observing zero structural failures out of roughly 300,000 particles tested under extreme conditions.

Despite this robust fuel integrity, the functional containment approach faces its most severe test during a steam generator tube rupture. This event creates a direct thermodynamic and chemical coupling between the secondary water system and the primary core. If isolation is delayed, water ingress drives graphite oxidation, rapid gas generation, and a severe pressure transient. To prevent the primary vessel from physically bursting the system must "exhale" this pressurized mixture from the reactor core through filtered vents to the environment. In this scenario, containment is no longer a hard barrier, but a managed release.

Debating Functional Containment: The Ag-110m Anomaly

While TRISO effectively traps the most catastrophic isotopes, it has a known chemical blind spot: **Silver-110m (Ag-110m)**. At normal operating temperatures above 1000°C, Ag-110m diffuses directly through the intact SiC layer, migrates into the bulk graphite, and chemically binds to the microscopic carbon dust continuously generated by the circulating pebbles.

If a steam ingestion accident forces the reactor to vent to the atmosphere, the high-velocity escaping gas will resuspend this highly radioactive carbon dust. This creates a sharp divide in safety engineering philosophy:

The "Pro" Argument: Mechanistic Source Term

- **Retention of the "Big Three":** Proponents argue that because TRISO securely retains most fission products such as Cesium (Cs-137) and Strontium (Sr-90), while significantly limiting (but not completely preventing) the release of noble gases like Krypton and Xenon, the fundamental goal of containment is met.
- **Filtered Venting:** The escaping gas is routed through massive HEPA and metallic blowdown filters. Designers argue these filters will catch the vast majority of the Ag-110m-laden dust.
- **Dose Limits:** Mathematical models suggest that the total inventory of escaped Ag-110m would remain below the EPA's Protective Action Guide (PAG) limits at the site boundary, making an expensive dome of sufficient size and strength to contain the core gases in the case of an accident financially unnecessary.

The "Con" Argument: Systemic and Environmental Risk

- **Filter Vulnerability:** Critics point out that during a violent, moisture-rich steam depressurization, filters can tear, clog, or be bypassed.
- **The 5 to 10-Year Plume:** Ag-110m is a powerful gamma emitter with a half-life of roughly 250 days. If a plume of contaminated dust escapes the filters, the downwind area will experience significantly elevated background radiation levels. Because it takes roughly 10 half-lives for an isotope to fully decay to background, the local environment (and biological food chains) would be impacted for 5 to 10 years.
- **Other Fission Products:** While under normal conditions intact TRISO particles have been shown to effectively trap Cesium and Strontium, off normal conditions (hot spots coupled with DFLOC - depressurized loss of forced cooling) could potentially compromise containment. Also, at least some of the particles will be defective, thus there will be some fission products in the primary loop coolant.
- **Deterministic vs. Probabilistic:** Opponents argue that relying on predictive computer models and filters to protect the public introduces an avoidable systemic risk, whereas a deterministic physical barrier for containing core gases under accident conditions provides guaranteed, passive protection against unforeseen modeling errors.

4 The Salt-Cooled Hybrid: Kairos Power (Hermes)

The Proponent: Google / TVA.

The Physics: A Fluoride Salt-Cooled High-Temperature Reactor (FHR) using TRISO pebbles in a liquid FLiBe salt coolant.

Current Status: Hermes 1 (non-power test) began nuclear safety-related construction in May 2025 in Oak Ridge, TN. Google has signed a landmark 500 MW Power Purchase Agreement with Kairos. The first deployment under this PPA will be the re-envisioned **Hermes 2** (recently scaled up to a **50 MWe** commercial demonstration plant), which will deliver power to the TVA grid in Oak Ridge to feed regional Google data centers.

4.1 Design & Physics

- **Atmospheric Pressure:** By using molten fluoride salt ($T_{boil} \approx 1430^\circ\text{C}$), Kairos eliminates the need for a thick pressure vessel. The reactor operates at 650°C but at near-ambient pressure, much like a Liquid Metal Fast Reactor (LMFR).
- **Buoyancy and Fuel:** Unlike a conventional HTGR where dense pebbles flow downward, Kairos' pebbles are designed to be **buoyant** in the salt. Pebbles are injected into the bottom of the vessel and float upward, allowing spent fuel to be harvested from the top of the pool.

4.2 Defense Against Steam: The Intermediate Loop

A major vulnerability of traditional gas-cooled reactors (such as the Xe-100) is water ingress from a steam generator rupture. Kairos solves this by physically separating the high-pressure steam cycle from the nuclear core:

- **Intermediate Salt Buffer:** Heat is transferred from the primary FLiBe loop to a non-radioactive **intermediate loop** filled with BeNaF (Sodium Fluoride-Beryllium Fluoride). The steam generator is attached to this secondary loop.
- **No Core Reactivity Risk:** If a high-pressure steam tube ruptures, steam blasts into the BeNaF loop, *not* the reactor core. This completely eliminates the risk of water acting as a positive moderator (reactivity spike) and prevents catastrophic steam-graphite chemical reactions in the core.

4.3 Containment Advantage over HTGRs

While Kairos also relies on the TRISO particle for "Functional Containment," its safety margins are vastly superior to a gas-cooled reactor during a primary boundary breach:

- **No Blowdown Force:** Because the primary loop is at atmospheric pressure, a pipe break will not result in a high-pressure blast. There is no thermodynamic driving force to push radiation out of the building or force the containment to "vent" to the atmosphere.
- **Chemical Retention:** Unlike helium (an inert gas that freely carries radioactive dust), molten FLiBe chemically binds to severe fission products like Cesium and Iodine. Even if a TRISO particle's shell fails, the salt chemically traps the radiation within the liquid pool.

4.4 The Constraints: Tritium, Freezing, and Reactions

- **Tritium (^3H) Production:** The Lithium in the FLiBe salt ($2\text{LiF} - \text{BeF}_2$) absorbs neutrons to produce Tritium. Tritium is notoriously "slippery" and permeates through hot metal heat exchangers. This is minimized (at a cost) by enriching the Lithium to 99.99% ^7Li . Kairos must still employ advanced capture systems to prevent Tritium from leaking into the steam cycle.
- **The "Freeze-Up" Risk:** If auxiliary heating fails, the salt "freezes" (solidifies) at $\approx 460^\circ\text{C}$, potentially trapping the buoyant fuel pebbles and damaging internal structural components.
- **Steam-Salt Hydrolysis (Non-Nuclear):** If a high-pressure steam generator tube ruptures, steam injects into the non-radioactive BeNaF intermediate loop. While it will not explode like liquid sodium, the superheated water chemically reacts with the salt (hydrolysis) to produce highly corrosive **Hydrofluoric Acid (HF)** gas and toxic **Beryllium Oxide (BeO)** sludge. The rapid expansion of steam can also flash-freeze the surrounding salt. This creates a severe chemical, pressurized, and operational mess, but leaves the nuclear core completely unaffected.

5 The Waste Trade-off: Volume vs. Intensity

A critical comparison for the "Nuclear Neighbor" discussion when siting reactors near data centers. It is important to note that the total fission product mass generated is strictly constant across all reactor designs on a MW_t basis; the engineering difference lies entirely in the *packaging* and the decay heat density.

- **BWRX-300 (Standard LWR Baseline):** Uses standard high-density UO_2 fuel pins. Because the water acts as the moderator and remains in the reactor, only the pure metal fuel is discharged. A standard gigawatt-scale LWR requires massive water pools for initial cooling, but ultimately only fills **1 to 2 dry casks per year per GWe**.
- **Xe-100 and Hermes (High Volume, Low Heat):** The TRISO pebble-bed design embeds microscopic fuel particles inside $\approx 200\text{g}$ graphite spheres. Because the graphite moderator is permanently bonded to the fuel, **the moderator must be discarded as High-Level Waste (HLW)**.
 - *The Logistical Penalty:* Because of the high burnup enabled by HALEU ($>160,000$ MWd/tHM), a 1 GWe HTGR discharges roughly **800,000 to 1 million spent pebbles annually**. While this is half the amount of older legacy designs, the massive graphite dilution still generates **≈ 10 to 15 dry casks per year per GWe** (roughly $5\times$ to $7\times$ the physical volume of a standard LWR).
 - *The Safety Advantage:* The resulting low power density means the decay heat of a single discharged pebble is negligible (roughly **8 Watts**). Spent fuel pools are entirely eliminated; pebbles drop straight from the reactor core into passively air-cooled, helium-filled dry casks with zero risk of meltdown.
- **Sodium (Low Volume, High Heat):** By utilizing HALEU in a liquid sodium fast spectrum, Sodium uses tightly packed metallic fuel pins to achieve massive burnup (>150 GWd/tHM). While the physical footprint of the spent fuel itself is incredibly dense, the extreme concentration of decay heat dictates the waste management strategy.

- *The Logistical Footprint*: Despite the higher burnup, the heavy metal mass (≈ 22 tHM per GWe-year) and intense decay heat mean a 1 GWe Sodium plant generates **1.5 to 2 dry casks per year**. This is virtually identical to the LWR baseline, because cask capacity is constrained by the ≈ 30 to 40 kW thermal limit rather than the physical size of the fuel assemblies.
- *The Cooling Penalty*: Because the spent metallic fuel is so intensely "hot" upon discharge, it entirely loses the immediate dry-storage advantage of gas-cooled reactors. Spent assemblies require mandatory active cooling and must be stored in internal racks submerged within the primary liquid sodium pool for years before they are thermally cool enough to be sealed in a dry cask.

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